| No. | Status | Date |
|---|---|---|
| HIS-2011-0207-DS-6-SG-102 | Issued | 2011-02-18 |

# UT Thickness Data Sheet - 6-SG-102

## Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 6-SG-102, located within the ATU-100. The scope of work involves the collection of thickness data to support the 5-year Class 1 survey as detailed in HIS-2011-0207. Kevin Doucette, acting as the lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., is responsible for the execution of the measurements and the recording of the data presented herein.

## Technical Basis and Procedures

All thickness measurements were performed in accordance with the requirements of ASTM E797. The field activities were conducted under the site-specific requirements of QSCP-NDT-001 Rev 2, which incorporates updated mandates for the CUI program, including insulation removal and inspection-port requirements. Furthermore, the inspection was performed under HIS-UT-002 Rev 1, the corporate ultrasonic thickness measurement procedure submitted by Halward Inspection Services Ltd. for QSCP acceptance.

## Component Design Data

The subject line, 6-SG-102, is a sour gas inlet constructed from A106 Gr.B. The nominal dimensions are 168.3 mm OD with a nominal wall thickness of 7.11 mm. The minimum allowable thickness (t-min) is 3.56 mm. The design parameters for this circuit are specified at a maximum allowable pressure of 1965 kPa(g) and a maximum allowable temperature of 66 ?C.

## Analysis and Survey Results

The data collected for 6-SG-102 is integrated into the broader survey findings of HIS-2011-0207. This survey identifies that the rich amine circuit is currently at half remaining life. Thickness readings are evaluated against the t-min of 3.56 mm. In accordance with site rules, any reading below this threshold is subject to an NCR. Remaining life and subsequent inspection intervals are determined by the larger of the long-term or short-term corrosion rates.

## Thickness readings, measured 2011-02-18

**Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-101-01 | 7.11 | 3.56 | 2006-02-24 | 6.59 | 6.39 | 0.049 | 0.040 | >50 | 2016-02-18 | |
| 6-SG-101-02 | 7.11 | 3.56 | 2006-02-24 | 6.51 | 6.14 | 0.061 | 0.074 | 34.7 | 2016-02-18 | |
| 6-SG-101-03 | 7.11 | 3.56 | 2006-02-24 | 6.42 | 6.17 | 0.061 | 0.050 | 42.9 | 2016-02-18 | |
| 6-SG-101-04 | 7.11 | 3.56 | 2006-02-24 | 6.28 | 6.05 | 0.070 | 0.046 | 35.8 | 2016-02-18 | |
| 6-SG-101-05 | 7.11 | 3.56 | 2006-02-24 | 6.22 | 5.87 | 0.085 | 0.070 | 27.2 | 2016-02-18 | |
| 6-SG-101-06 | 7.11 | 3.56 | 2006-02-24 | 6.49 | 6.27 | 0.054 | 0.044 | >50 | 2016-02-18 | |
| 6-SG-101-08 | 7.11 | 3.56 | — | — | 6.33 | — | — | — | 2016-02-18 | |

**Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-102-01 | 7.11 | 3.56 | 2006-02-24 | 6.26 | 5.75 | 0.093 | 0.102 | 21.4 | 2016-02-18 | |
| 6-SG-102-02 | 7.11 | 3.56 | 2006-02-24 | 6.25 | 5.80 | 0.088 | 0.090 | 24.8 | 2016-02-18 | |
| 6-SG-102-03 | 7.11 | 3.56 | 2006-02-24 | 6.26 | 5.80 | 0.083 | 0.092 | 24.3 | 2016-02-18 | |
| 6-SG-102-04 | 7.11 | 3.56 | 2006-02-24 | 6.22 | 5.73 | 0.084 | 0.098 | 22.1 | 2016-02-18 | |
| 6-SG-102-05 | 7.11 | 3.56 | 2006-02-24 | 6.72 | 6.46 | 0.045 | 0.052 | >50 | 2016-02-18 | |
| 6-SG-102-06 | 7.11 | 3.56 | — | — | 6.41 | — | — | — | 2016-02-18 | |

**Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-01 | 8.56 | 3.81 | 2006-02-24 | 8.27 | 8.32 | 0.013 | -0.010 | >50 | 2021-02-18 | |
| 4-RA-101-02 | 8.56 | 3.81 | 2006-02-24 | 8.27 | 8.20 | 0.023 | 0.014 | >50 | 2021-02-18 | |
| 4-RA-101-03 | 8.56 | 3.81 | 2006-02-24 | 8.10 | 7.98 | 0.042 | 0.024 | >50 | 2021-02-18 | |
| 4-RA-101-04 | 8.56 | 3.81 | 2006-02-24 | 8.34 | 8.21 | 0.028 | 0.026 | >50 | 2021-02-18 | |
| 4-RA-101-05 | 8.56 | 3.81 | 2006-02-24 | 8.29 | 8.16 | 0.029 | 0.026 | >50 | 2021-02-18 | |
| 4-RA-101-07 | 8.56 | 3.81 | 2006-02-24 | 8.30 | 8.30 | 0.020 | 0.000 | >50 | 2021-02-18 | |
| 4-RA-101-08 | 8.56 | 3.81 | 2006-02-24 | 8.07 | 8.02 | 0.044 | 0.010 | >50 | 2021-02-18 | |
| 4-RA-101-09 | 8.56 | 3.81 | 2006-02-24 | 8.43 | 8.32 | 0.017 | 0.022 | >50 | 2021-02-18 | |
| 4-RA-101-10 | 8.56 | 3.81 | 2006-02-24 | 8.27 | 8.28 | 0.026 | -0.002 | >50 | 2021-02-18 | |
| 4-RA-101-11 | 8.56 | 3.81 | 2006-02-24 | 8.14 | 8.04 | 0.043 | 0.020 | >50 | 2021-02-18 | |

**Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-102-01 | 8.56 | 3.81 | 2006-02-24 | 8.41 | 8.33 | 0.014 | 0.016 | >50 | 2021-02-18 | |
| 4-RA-102-02 | 8.56 | 3.81 | 2006-02-24 | 8.15 | 7.99 | 0.039 | 0.032 | >50 | 2021-02-18 | |
| 4-RA-102-03 | 8.56 | 3.81 | 2006-02-24 | 8.27 | 8.04 | 0.035 | 0.046 | >50 | 2021-02-18 | |
| 4-RA-102-04 | 8.56 | 3.81 | 2006-02-24 | 8.24 | 7.96 | 0.039 | 0.056 | >50 | 2021-02-18 | |
| 4-RA-102-05 | 8.56 | 3.81 | 2006-02-24 | 8.19 | 8.07 | 0.032 | 0.024 | >50 | 2021-02-18 | |
| 4-RA-102-06 | 8.56 | 3.81 | 2006-02-24 | 8.15 | 8.01 | 0.033 | 0.028 | >50 | 2021-02-18 | |
| 4-RA-102-07 | 8.56 | 3.81 | 2006-02-24 | 8.29 | 8.10 | 0.032 | 0.038 | >50 | 2021-02-18 | |
| 4-RA-102-08 | 8.56 | 3.81 | 2006-02-24 | 8.05 | 7.88 | 0.041 | 0.034 | >50 | 2021-02-18 | |
| 4-RA-102-09 | 8.56 | 3.81 | 2006-02-24 | 8.18 | 7.98 | 0.038 | 0.040 | >50 | 2021-02-18 | |
| 4-RA-102-11 | 8.56 | 3.81 | — | — | 8.19 | — | — | — | 2021-02-18 | |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-01 | 10.97 | 4.57 | 2006-02-24 | 10.76 | 10.47 | 0.035 | 0.058 | >50 | 2021-02-18 | |
| 6-RA-103-02 | 10.97 | 4.57 | 2006-02-24 | 10.50 | 10.23 | 0.058 | 0.054 | >50 | 2021-02-18 | |
| 6-RA-103-03 | 10.97 | 4.57 | 2006-02-24 | 10.67 | 10.42 | 0.038 | 0.050 | >50 | 2021-02-18 | |
| 6-RA-103-04 | 10.97 | 4.57 | 2006-02-24 | 10.40 | 9.97 | 0.068 | 0.086 | >50 | 2021-02-18 | |
| 6-RA-103-05 | 10.97 | 4.57 | 2006-02-24 | 10.34 | 10.10 | 0.057 | 0.048 | >50 | 2021-02-18 | |
| 6-RA-103-06 | 10.97 | 4.57 | 2006-02-24 | 10.74 | 10.56 | 0.030 | 0.036 | >50 | 2021-02-18 | |
| 6-RA-103-07 | 10.97 | 4.57 | 2006-02-24 | 10.66 | 10.55 | 0.029 | 0.022 | >50 | 2021-02-18 | |
| 6-RA-103-08 | 10.97 | 4.57 | 2006-02-24 | 10.58 | 10.44 | 0.035 | 0.028 | >50 | 2021-02-18 | |
| 6-RA-103-09 | 10.97 | 4.57 | 2006-02-24 | 10.67 | 10.56 | 0.030 | 0.022 | >50 | 2021-02-18 | |
| 6-RA-103-10 | 10.97 | 4.57 | 2006-02-24 | 10.27 | 10.05 | 0.065 | 0.044 | >50 | 2021-02-18 | |
| 6-RA-103-13 | 10.97 | 4.57 | — | — | 10.45 | — | — | — | 2021-02-18 | |

**Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-104-01 | 8.56 | 3.81 | 2006-02-24 | 8.16 | 7.95 | 0.037 | 0.042 | >50 | 2021-02-18 | |
| 4-RA-104-02 | 8.56 | 3.81 | 2006-02-24 | 8.23 | 8.14 | 0.027 | 0.018 | >50 | 2021-02-18 | |
| 4-RA-104-03 | 8.56 | 3.81 | 2010-09-13 | 8.59 | 8.56 | 0.069 | 0.069 | >50 | 2021-02-18 | |
| 4-RA-104-04 | 8.56 | 3.81 | 2010-09-13 | 8.52 | 8.60 | -0.185 | -0.185 | >50 | 2021-02-18 | |
| 4-RA-104-05 | 8.56 | 3.81 | 2010-09-13 | 8.43 | 8.57 | -0.324 | -0.324 | >50 | 2021-02-18 | |
| 4-RA-104-06 | 8.56 | 3.81 | 2006-02-24 | 8.09 | 7.98 | 0.041 | 0.022 | >50 | 2021-02-18 | |
| 4-RA-104-07 | 8.56 | 3.81 | 2006-02-24 | 7.97 | 7.78 | 0.054 | 0.038 | >50 | 2021-02-18 | |
| 4-RA-104-08 | 8.56 | 3.81 | — | — | 8.18 | — | — | — | 2021-02-18 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Kevin Doucette | lead UT technician, SNT-TC-1A UT Level II | 2011-02-18 | _____ |